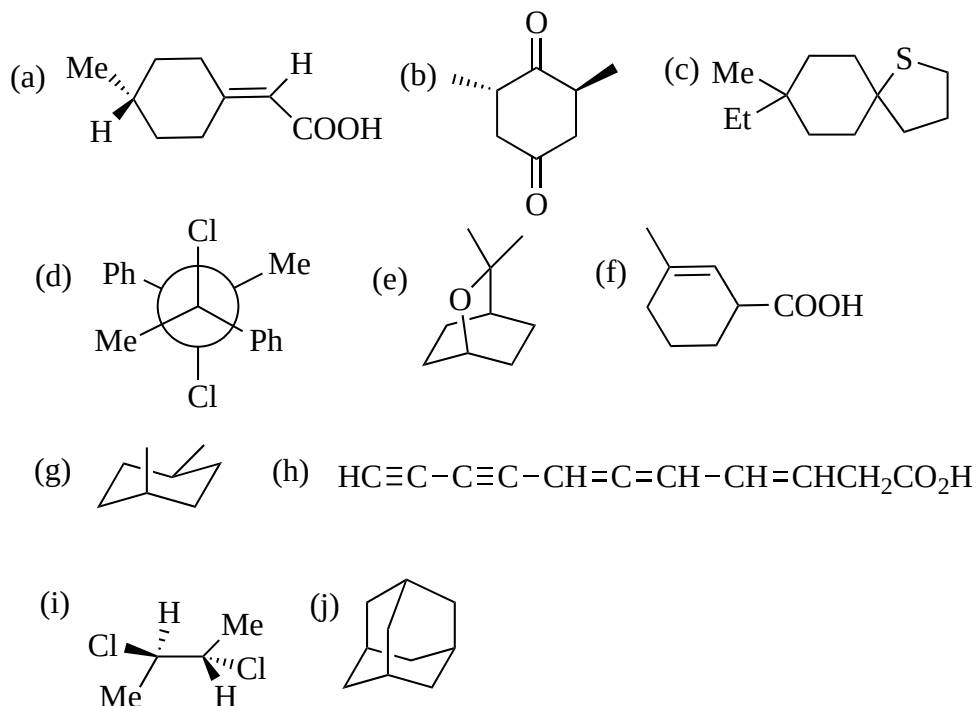


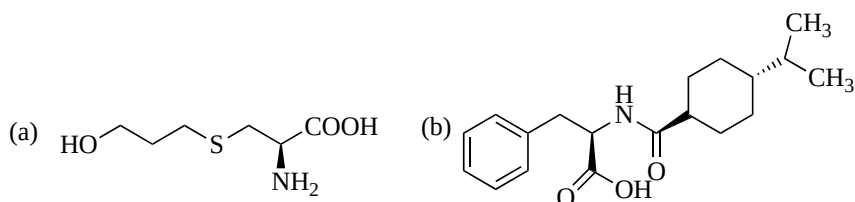
ACML 101: Organic Chemistry: Assignment 1

Instructor: Dr. Nidhi Jain

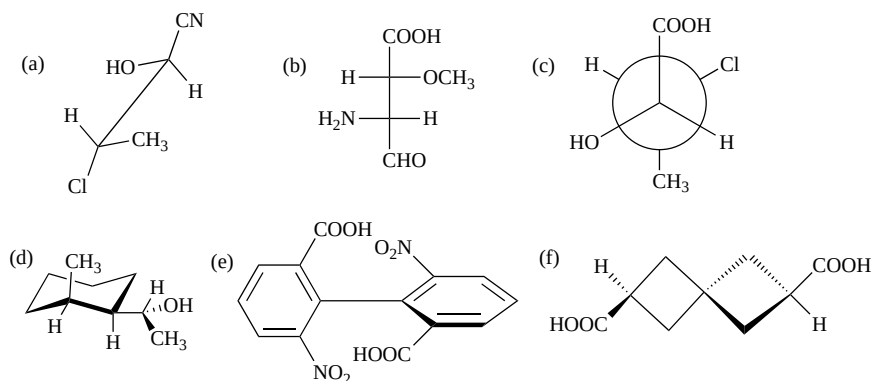
1. Identify the following molecules as chiral or achiral?



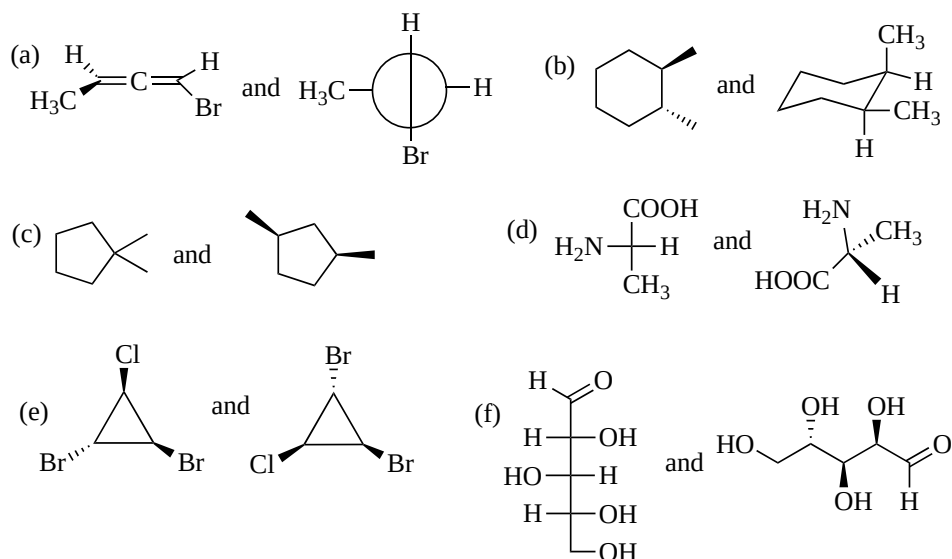
2. Assign R/S configuration to the following compounds



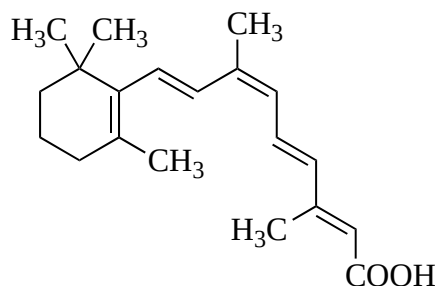
3. Determine the absolute configuration of the following compounds



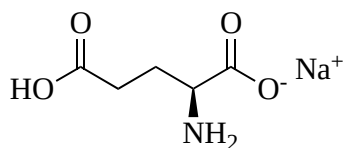
4. What is the relationship between the following pair of compounds?



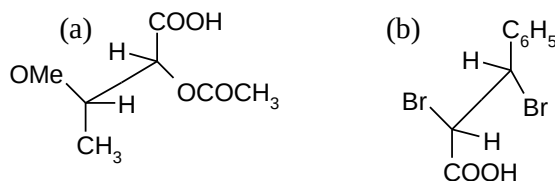
5. Assign E/Z configuration to the double bonds in the compound given below. How many stereo isomers are possible?



6. Pure MSG, a common flavor enhancer, exhibits a specific rotation of +24. (a) Calculate the *ee* of a solution whose $[\alpha]$ is +10. (b) If the *ee* of a solution is 80 %, what is the $[\alpha]$ for this solution?



7. Draw the following molecules in their respective Fischer projection formula



8. Draw the Wedge, Fischer, Newman, and Sawhorse projections for the **(2S,3S)-3-bromobutan-2-amine**.